Intel 80386 and Pentium Microprocessor

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Intel 80386

- The 80386 family of microprocessors of Intel Corporation is the first 32 bit version of the 8086 family-a switch from 16 bit to 32 bit
- 80386 has upward compatibility with 8086,8088,80286 etc
- The 80386 was launched in October 1985, but full-function chips were first delivered in the third quarter of 1986
- Although it had long been obsolete as a personal computer CPU, Intel and others had continued making the chip for embedded systems.
- Used in aerospace technology

Versions of 80386

80386DX – the full version

- The first member in 80386 family
- this CPU could work with 16-bit and 32-bit external buses.
- Comprises of both 32-bit internal registers and 32-bit external bus.

80386SX –the reduced bus version

- low cost version of the 80386.
- This processor had 16 bit external data bus,32-bit internal registers and 24-bit external address bus.

80386SL –

- low-power microprocessor with power management features, with 16-bit external data bus and 24-bit external address bus.
- The processor included ISA bus controller, memory controller and cache controller.

Embedded 80376 and 80386EX processors.- Still in use today.

FEATURES OF 80386:

Two versions of 80386 are commonly available:

1) 80386DX

2)80386SX

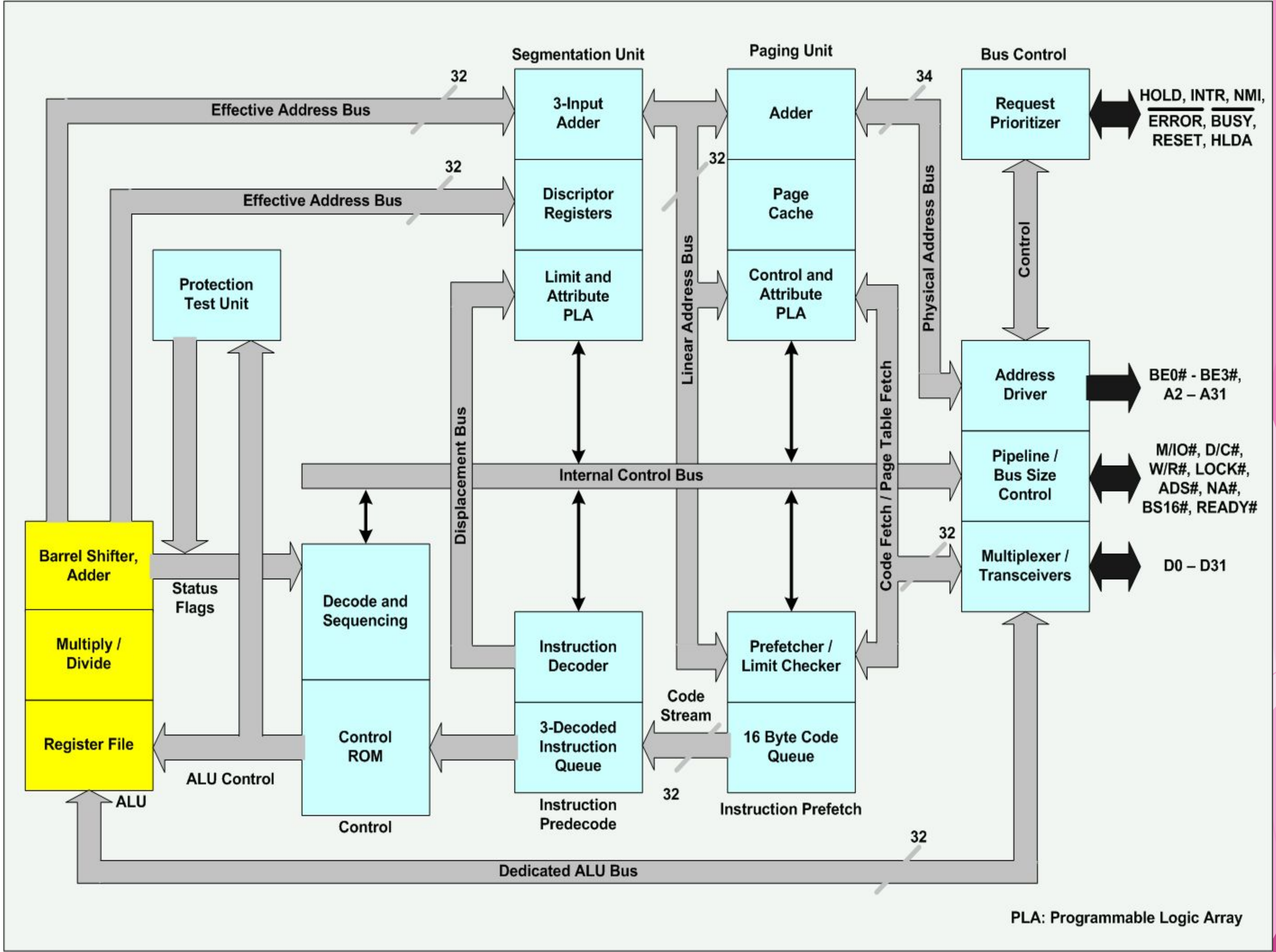
80386DX	80386SX
1) 32 bit address bus 32bit data bus	1) 24 bit address bus 16 bit data bus
1) Packaged in 132 pin ceramic pin grid array(PGA)	2) 100 pin flat package
3) Address 4GB of memory	3) 16 MB of memory

80386SX was developed after the DX for application that didn’t require the full 32-bit bus version. It is found in many PCs use the same basic mother board design as the 80286. Most application less than the 16MB of memory, so the SX is popular and less costly version of the 80386 microprocessor.

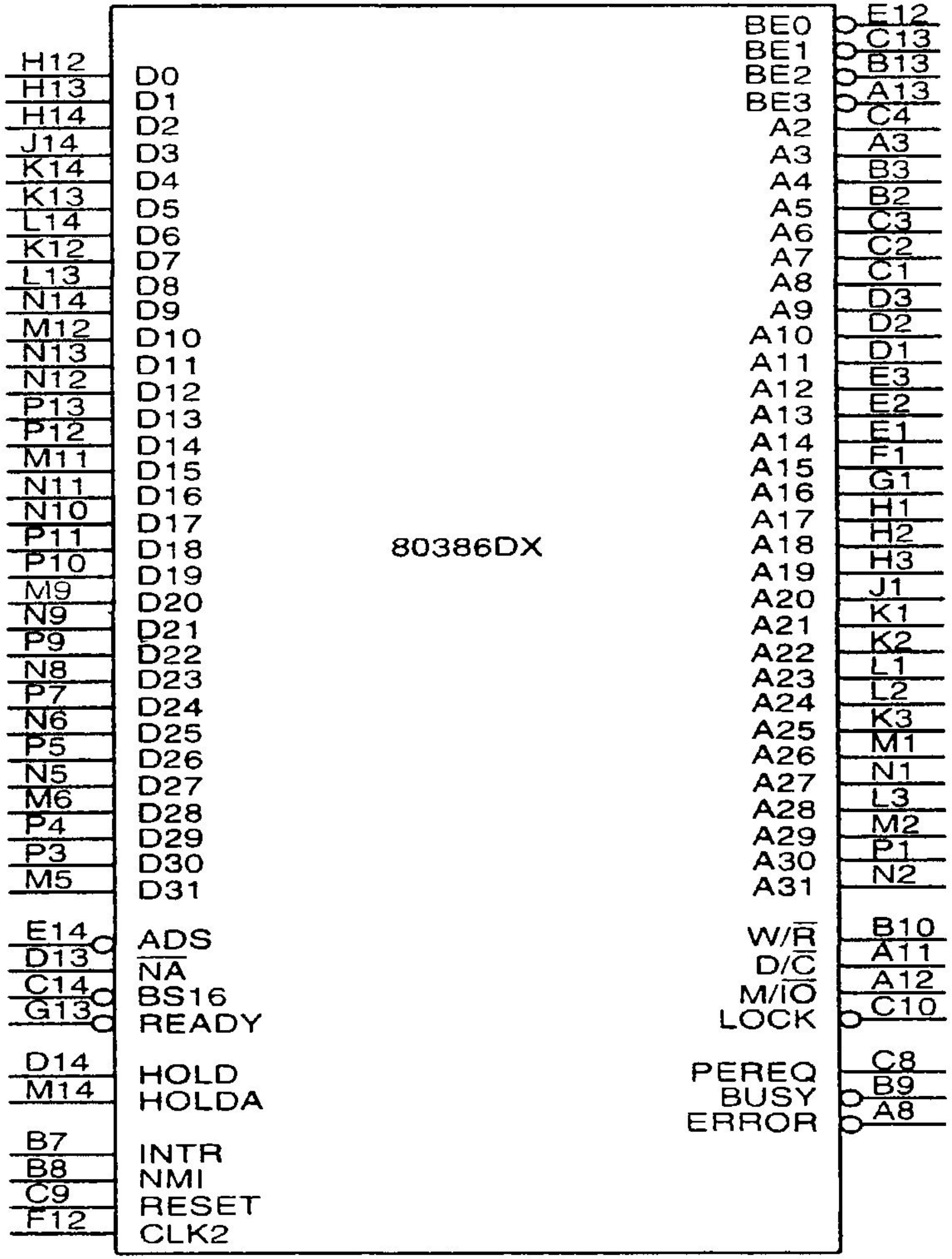
- **The 80386 CPU supports 16k no:** of segments and thus total virtual memory space is 4GB *16 k=64 terabytes
- **Memory management section supports**
  - Virtual memory
  - Paging
  - 4 levels of protection
- **20-33 MHz frequency**

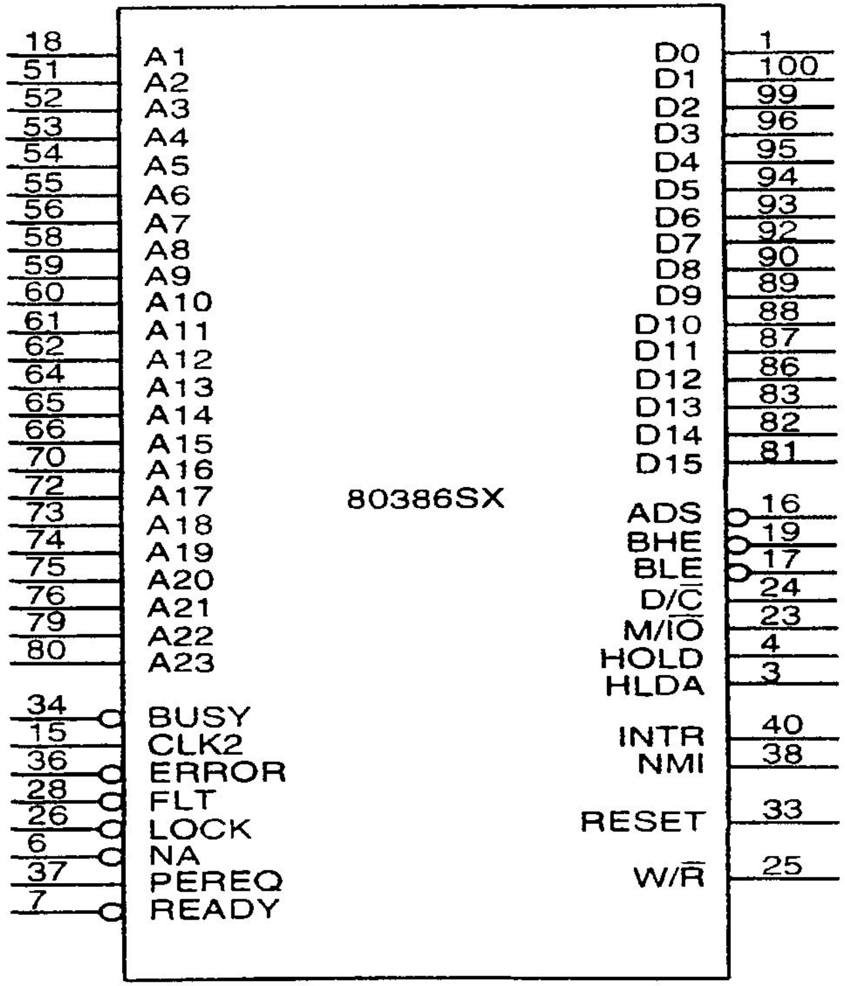
Internal Architecture

- To enhance performance 80386 has 6 functional units, processing in parallel:
  - The bus unit
  - The prefetch unit
  - The decode unit
  - The execution unit
  - The page unit
  - The segment unit



Segmentation Unit
Paging Unit
Bus Control
32
Effective Address Bus
3-Input Adder
Discriptor Registers
Limit and Attribute PLA
Adder
Page Cache
Control and Attribute PLA
34
32
Linear Address Bus
Physical Address Bus
Request Prioritizer
HOLD, INTR, NMI, ERROR, BUSY, RESET, HLDA
Control
32
Effective Address Bus
Protection Test Unit
Displacement Bus
Address Driver
BE0# - BE3#, A2 – A31
Internal Control Bus
Code Fetch / Page Table Fetch
Pipeline / Bus Size Control
M/IO#, D/C#, W/R#, LOCK#, ADS#, NA#, BS16#, READY#
32
Multiplexer / Transceivers
D0 – D31
Barrel Shifter, Adder
Status Flags
Decode and Sequencing
Multiply / Divide
Instruction Decoder
Prefetcher / Limit Checker
Code Stream
Register File
Control ROM
3-Decoded Instruction Queue
16 Byte Code Queue
ALU Control
ALU
32
Control
Instruction Predecode
Instruction Prefetch
32
Dedicated ALU Bus
PLA: Programmable Logic Array


80386DX
H12 D0
H13 D1
H14 D2
J14 D3
K14 D4
K13 D5
L14 D6
K12 D7
L13 D8
N14 D9
M12 D10
N13 D11
N12 D12
P13 D13
P12 D14
M11 D15
N11 D16
N10 D17
P11 D18
P10 D19
M9 D20
N9 D21
P9 D22
N8 D23
P7 D24
N6 D25
P5 D26
N5 D27
M6 D28
P4 D29
P3 D30
M5 D31
E14 ADS
D13 NA
C14 BS16
G13 READY
D14 HOLD
M14 HOLDA
B7 INTR
B8 NMI
C9 RESET
F12 CLK2
BE0 E12
BE1 C13
BE2 B13
BE3 A13
A2 C4
A3 A3
A4 B3
A5 B2
A6 C3
A7 C2
A8 C1
A9 D3
A10 D2
A11 D1
A12 E3
A13 E2
A14 E1
A15 F1
A16 G1
A17 H1
A18 H2
A19 H3
A20 J1
A21 K1
A22 K2
A23 L1
A24 L2
A25 K3
A26 M1
A27 N1
A28 L3
A29 M2
A30 P1
A31 N2
W/R B10
D/C A11
M/IO A12
LOCK C10
PEREQ C8
BUSY B9
ERROR A8


80386SX
18 A1
51 A2
52 A3
53 A4
54 A5
55 A6
56 A7
58 A8
59 A9
60 A10
61 A11
62 A12
64 A13
65 A14
66 A15
70 A16
72 A17
73 A18
74 A19
75 A20
76 A21
79 A22
80 A23
34 BUSY
15 CLK2
36 ERROR
28 FLT
26 LOCK
6 NA
37 PEREQ
7 READY
D0 1
D1 100
D2 99
D3 96
D4 95
D5 94
D6 93
D7 92
D8 90
D9 89
D10 88
D11 87
D12 86
D13 83
D14 82
D15 81
ADS 16
BHE 19
BLE 17
D/C 24
M/IO 23
HOLD 4
HLDA 3
INTR 40
NMI 38
RESET 33
W/R 25

The Bus Unit

- The bus unit is the interface to the external devices.
- The bus interface unit provides a 32-bit data bus, a 32-bit address bus and the signals needed to control transfers over the bus.
- In fact, 8-bit, 16-bit and 32-bit data transfers are supported.
- 80386 has separate pins for its address and data bus lines.
- This processing unit contains the latches and drivers for the address bus, transceivers for the data bus, and control logic for signaling whether a memory input/output, or interrupt-acknowledgement bus cycle is to be performed.

The Prefetch Unit

- The prefetch unit performs a mechanism known as an *instruction stream queue*.
- This queue permits a prefetch upto 16 bytes of instruction code.
- Whenever the queue is not full, and the execution unit is not asking the bus unit to read or write data from the memory, the prefetch queue supplies addresses to the bus interface unit and signals it to look ahead in the program by fetching the next sequential instructions.
- Prefetched instructions are held in the FIFO queue for use by the instruction decoder.
- Whenever bytes are loaded into the input end of the queue, they are automatically shifted up through the FIFO to the empty location near the output.

The Decode Unit

- The decode unit accesses the output end of the prefetch unit's instruction queue.
- It reads the machine-code instructions from the output side of the prefetch queue and decodes them into microcode instruction format used by the execution unit, thus it off-loads the responsibility for the instruction decoding from the execution unit.
- The instruction queue, a part of the decode unit permits three fully decoded instructions to be held waiting for use by the execution unit.
- Thus it improves the performance of the CPU.

The Execution Unit

- The execution unit involves the arithmetic/logic unit-ALU, registers, special multiply, divide, and shift hardware, and a control ROM.
- The control ROM contains the microcode sequences that define the operation performed by each of the machine code instructions.
- The execution unit reads the decoded instructions from the instruction queue and performs the operations that are specified.
- During the execution of an instruction, it requests the segment and page units to generate operand addresses and the bus interface unit to perform read or write bus cycles to access data in memory or I/O devices.

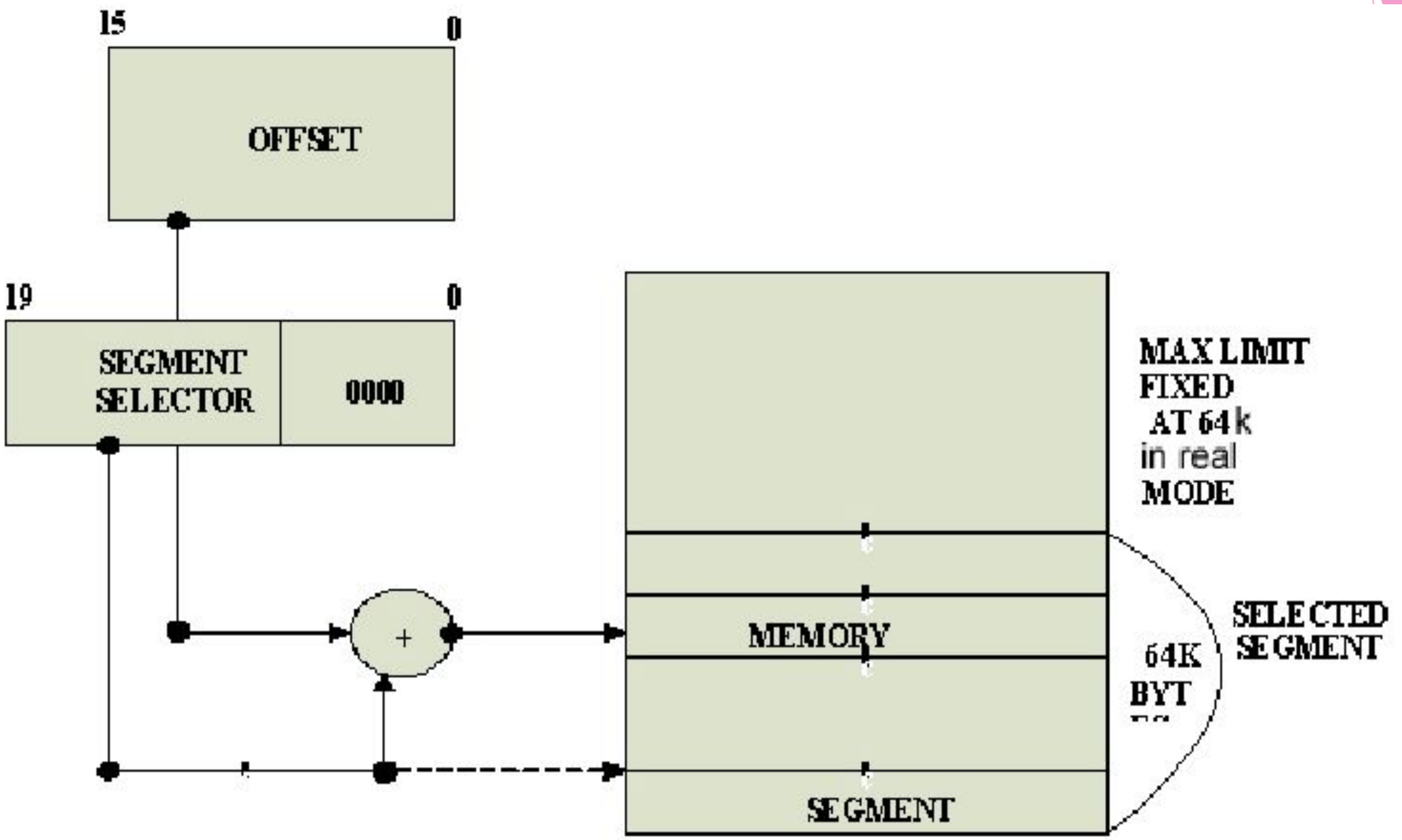
The Page and Segment Unit

- The segment and the Page units provide the memory management and protection services for the 80386.

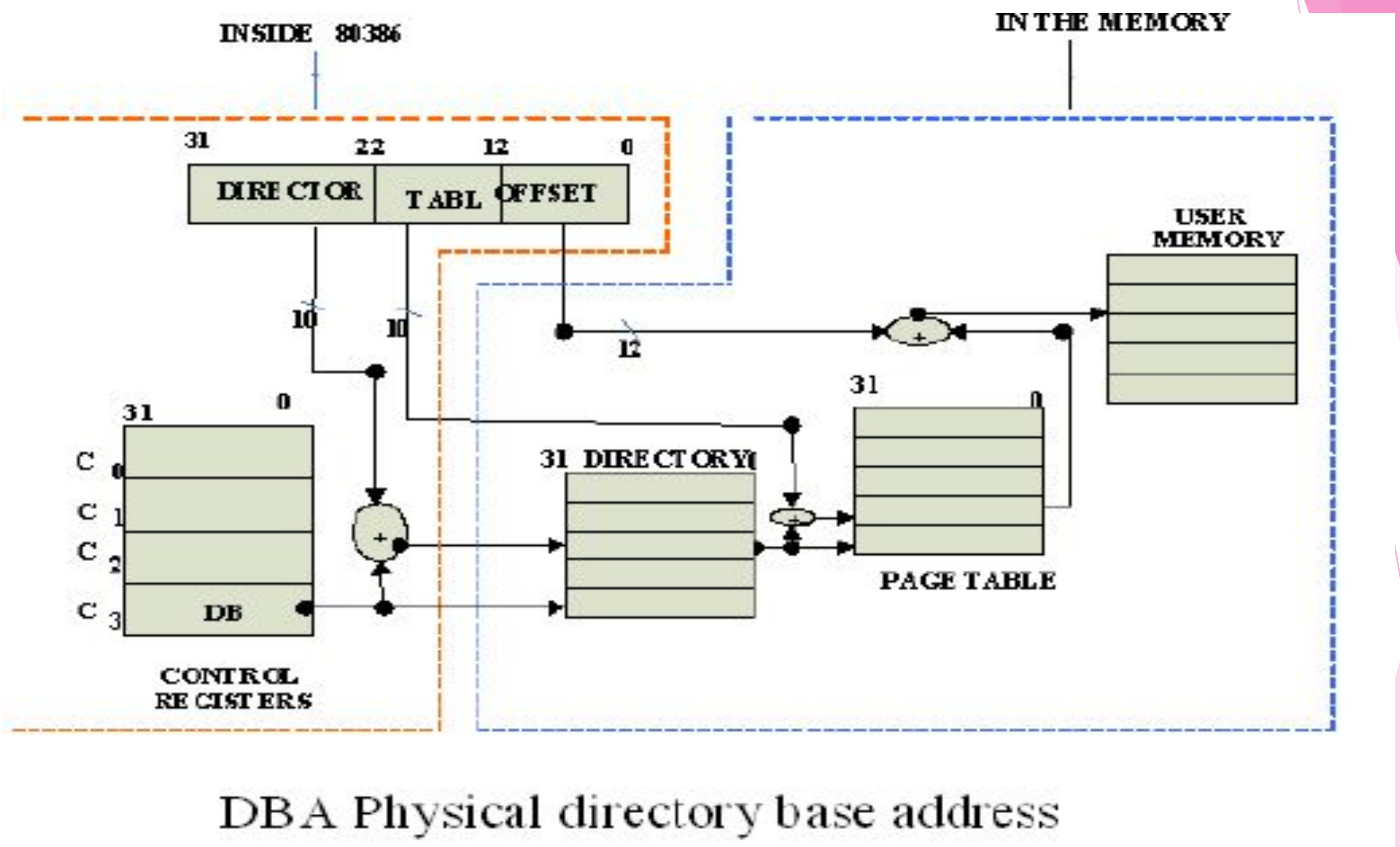
- They offload the responsibility for address generation, address translation and segment checking from the bus interface unit and thereby further boosting the performance of the CPU.

- The segment unit implements the segmentation model of the 386 memory management.

- It contains dedicated hardware for performing high speed address calculations, logical to linear address translation and protection checks.



Physical Address Formation In Real Mode Of 80386



Virtual 8086 Mode

- In its protected mode of operation, 80386DX provides a virtual 8086 operating environment to execute the 8086 programs.
- The real mode can also used to execute the 8086 programs along with the capabilities of 80386, like protection and a few additional instructions.
- Once the 80386 enters the protected mode from the real mode, it cannot return back to the real mode without a reset operation.
- Thus, the virtual 8086 mode of operation of 80386, offers an advantage of executing 8086 programs while in protected mode.
- The address forming mechanism in virtual 8086 mode is exactly identical with that of 8086 real mode.

Pentium microprocessors

- A 32-bit microprocessor introduced by Intel in 1993.
- It contains 3.3 million transistors, nearly triple the number contained in its predecessor, the 80486 chip.
- The Pentium processor has been superseded by the Pentium Pro and Pentium II microprocessors.
- Since 1993, Intel has developed the Pentium III and more recently the Pentium 4 microprocessors.

Pentium microprocessors - P-III

- Intel builds on the technology it developed with the Pentium II microprocessors.
- The Pentium III processor comes with a Synchronized Dynamic Random Access Memory (SDRAM), allowing for an extremely fast transfer of data between the microprocessor and the memory.
- 70 new instructions, called Streaming SIMD Extensions, enhance multimedia and 3D performance.
- An advanced transfer cache and system buffering are able to meet higher data bandwidth requirements.

Pentium microprocessors - P-III

- Launched February 1999 - Available in speed levels of 450, 500, 550, and 600MHz
- .25 Micron Manufacturing Process
- 32KB of Level 1 Cache (operating at CPU's full core speed)
- 512KB of Level 2 Cache (operating at ½ of CPU's core speed)
- 100 MHz front-side bus speed
- MMX support
- The branch prediction/recovery pipeline was doubled to include 10-stages from the P-II.

Pentium microprocessors - P-4

- The next generation of microprocessors from Intel.
- Pentium 4 is the product of a serious redesign.
- The move from Pentium II to Pentium III added two million transistors.
- The Pentium 4 sports a whopping 42 million - 14 million more than the currently available Pentium III Coppermine processors. (Actually, 55 million for redundancy/reliability.)

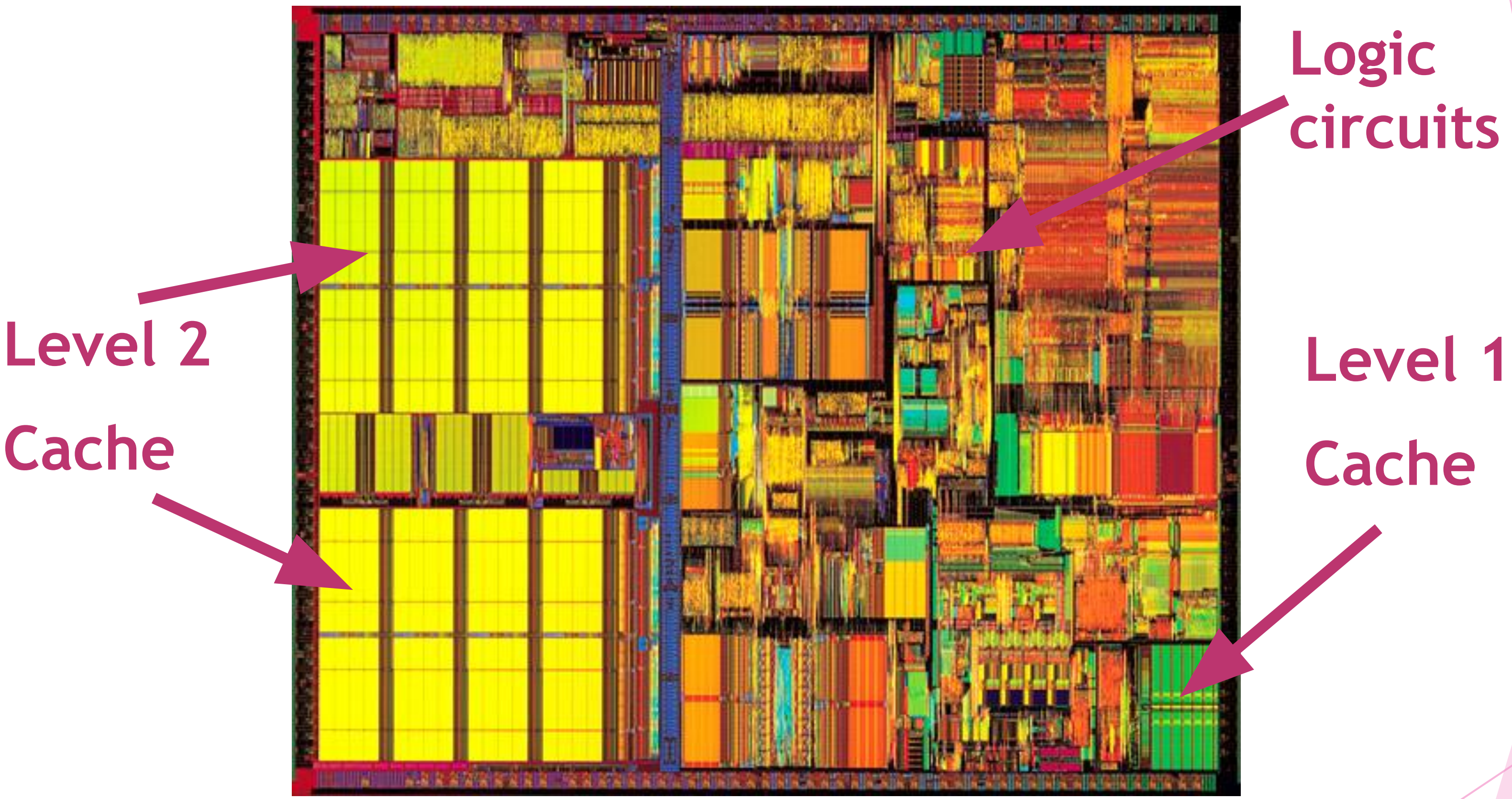
Pentium microprocessors - P-4

- The Pentium 4 is significantly larger than its predecessor.
- The P4 was first etched using the same .18-micron, aluminum trace process as the P-III Coppermine.
- The current P4 etching is the .13-micron process
- Now that the Pentium 4 is upon us the pipeline depth has been doubled once again to 20 stages - NetBurst microarchitecture .
- By doubling the depth of the branch prediction pipe, the penalty associated with miss-predictions is greatly increased.
- This results in a lower average number of instructions successfully executed per clock cycle.

Pentium microprocessors - P-4

- To compensate for the lower IPC, Intel has implemented:
  - **Execution Trace Cache**
  - **Dynamic Execution Engine**
- A 32-bit microprocessor, hyper-pipelined technology, a rapid execution engine and a quad-pumped 100MHz-system bus, delivering the equivalent of 3.2GB/s of bandwidth- three times the bandwidth of the Pentium III
- It transfers data at the equivalent of 400MHz bus speed

Pentium microprocessors

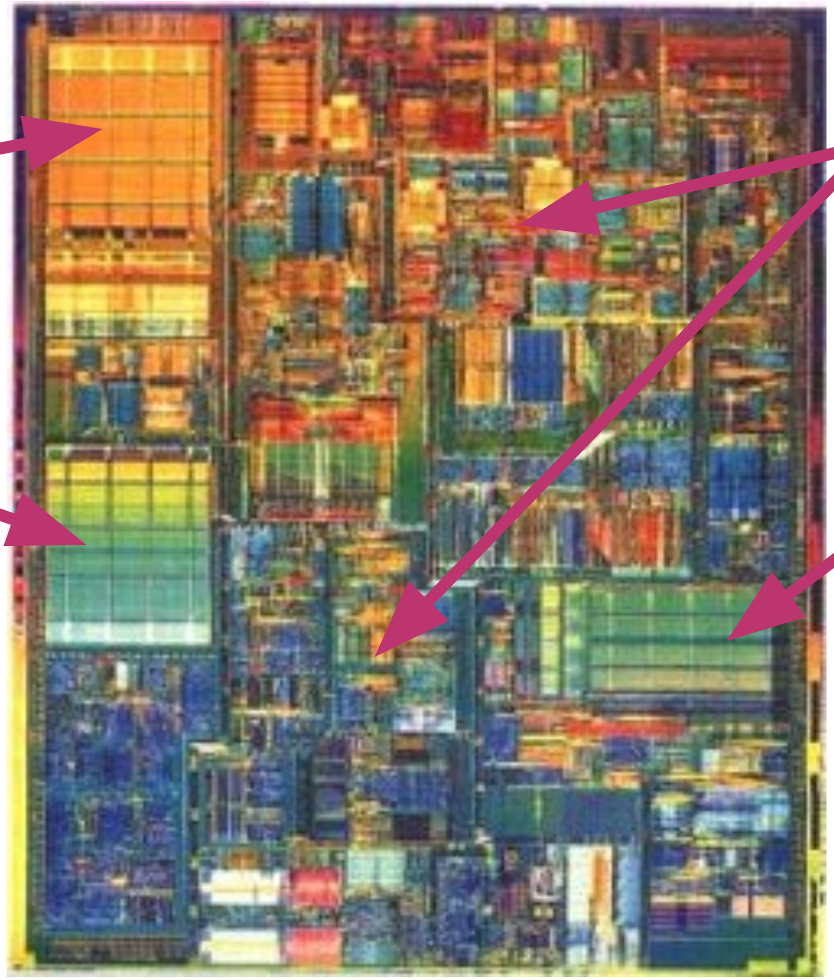


Pentium III

Pentium microprocessors

Level 2

Cache



Increased logic circuits

Cache

Level 1

Cache